

WORKSHEET - NAPLAN Year 9

Number Rates & Ratios

Teacher: Niné Labuschagne

Exam Equivalent Time: 13.75 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. Number, NAP-61529

Shelly and Carly collect dolls.

The ratio of the number of dolls Shelly owns compared to Carly is 3 : 2.

Shelley owns 12 dolls.

How many dolls does Carly own?

2 6 8 18
☐ ☐ ☐ ☐

2. Number, NAP-79138

In a hospital, there are 40 nurses and 600 patients.

What is the ratio of nurses to patients?

2:3 1:15 30:2 1:16
☐ ☐ ☐ ☐

3. Number, NAP-96744

The colour orange can be made by mixing 5 litres of yellow paint with 8 litres of red paint.

If 140 litres of yellow paint are used in this mixture, how much red paint is needed?

18 litres 28 litres 88 litres 224 litres
☐ ☐ ☐ ☐

4. Number, NAP-02683

Brendan lives in New York and returns to visit his hometown, Perth, every three years.

His next visit in 2020 will also let him attend the Mining Expo that is held once every two years.

When was the last year that Brendan visited Perth at the same time as the Mining Expo was held?

2014 2015 2016 2017
☐ ☐ ☐ ☐

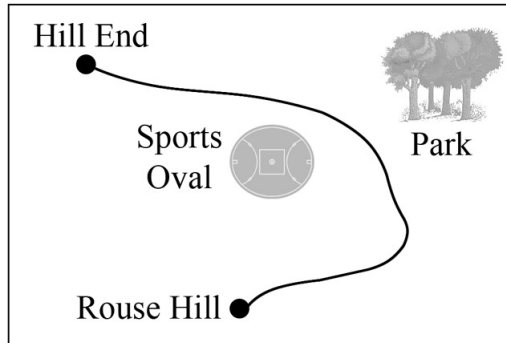
5. Number, NAP-02685

A christmas pudding recipe uses 3 cups of sugar for every 4 cups of sultanas.

Select the correct combination of sugar and sultanas for this recipe.

☐ $\frac{1}{3}$ cup of sugar, $\frac{1}{4}$ cup of sultanas
☐ 1 cup of sugar, 1.5 cups of sultanas
☐ 1.5 cups of sugar, 2 cups of sultanas
☐ 4 cups of sugar, 5 cups of sultanas

6. Geometry, NAP-14547



Scale : 1 cm = 1500 m

The distance on the map from Hill End to Rouse Hill along the road is 5 cm.

How long will it take to walk from Hill End to Rouse Hill along the road at a speed of 6 kilometres per hour?

- ☐ 1 hour 15 minutes
- ☐ 1 hour 25 minutes
- ☐ 12 hours 5 minutes
- ☐ 12 hours 30 minutes

7. Number, NAP-79167

Paul has a paddock that takes him 56 steps to walk across.

One of his steps is 80 centimetres long.

His younger sister, Caroline, has a step length of 70 centimetres.

How many steps does it take Caroline to walk across the same paddock?

- 49** **56** **64** **80** **100**
- ☐ ☐ ☐ ☐ ☐

8. Number, NAP-79196

Scientists estimate that in 1 km² of swamp, 30 000 mosquito eggs are laid every minute.
Which value is the closest to the number of mosquito eggs that would be laid in one day?

- 7.2×10^4 7.2×10^5 4.32×10^5 4.32×10^7
- ☐ ☐ ☐ ☐

9. Number, NAP-79194

There are 48 people in a meeting room.

The ratio of females to males in the room is 13 to 11.

How many females need to leave the room so that the ratio of females to males in the room is 1 to 1?

10. Number, NAP-08307

Saul's workshop requires lighting for 30 hours per week, every week of the year.

He uses globes that need replacing after 10 000 hours of use.

After how long, to the nearest year, would Saul need to replace any individual globe?

 years

11. Number, NAP-55673

It takes 15 minutes for all of the sand to flow through a sand timer.

The sand timer contains 475 grams of sand.

Each millilitre of sand weighs 1.5 grams.



How many millilitres of sand flow through the timer per minute?

10.3

21.1

31.7

47.5

☐☐☐☐